Meiosis consists of two divisions. They are:

- **Heterotypic Division** or First Meiotic Division
- **Homotypic Division** or Second Meiotic Division

A. Heterotypic division

It divides the diploid cell into two haploid cells. The daughter cells resulting from this division are different from the parent cell in the chromosome number (Heterotypic). This consists of 5 stages:

a. Prophase I
b. Metaphase I
c. Anaphase I
d. Telophase I
e. Cytokinesis I

a. Prophase I

Prophase I takes a longer duration and is sub divided into five stages. They are: Leptotene, Zygotene, Pachytene, Diplotene and Diakinesis.

Leptotene: The chromosomes become uncoiled and assume long thread like structures and take up a specific orientation inside the nucleus. They form a **bouquet stage**.

Zygotene (Zygon-adjoining): Two homologous chromosomes approach each other and begin to pair. Pairing of homologous chromosomes is called as **synapsis**.

Pachytene (Pachus-thick): The chromosomes are visible as long paired twisted threads. The pairs so formed are called **bivalents**. Each bivalent now contains four chromatids (**tetrad stage**). Homologous chromosomes of each pair begin to separate. They do not completely separate, but remain attached together at one or more points by X- shaped arrangements known as **chiasmata**. The chromatids break at these points and the broken segments may be interchanged (crossing over). As a result, the **genetic recombination** takes place.

Diplotene: Each individual chromosome of each bivalent begins to split longitudinally into two similar chromatids. The homologous chromosomes repel each other and separate. Chiasmata begin to move along the length of the chromosome from the centromere towards the end resulting in **terminalization**.

Diakinesis: The paired chromosomes are shortened and thickened. The nuclear membrane and nucleolus begin to disappear. Spindle fibres make their appearance.

b. Metaphase I

The chromosomes move towards the equator and finally they orient themselves on the equator. The two chromatids of each chromosome do not separate. The centromere does not divide.

c. Anaphase I

Each homologous chromosome with its two chromatids and undivided centromere move towards the opposite poles of the cell. This stage of the chromosome is called **Diad**.

d. Telophase I

The haploid number of chromosomes after reaching their respective poles become uncoiled

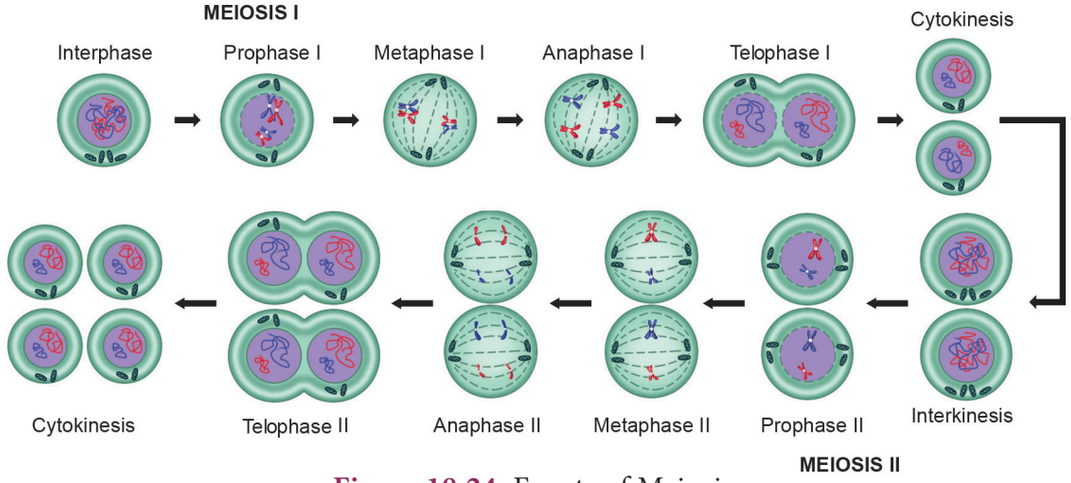


Figure 18.24 Events of Meiosis

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